Durability

- **Durability** is the wood's ability to resist decay from exposure to the elements and attack from insects.
- Hardwoods such as teak are the most durable woods and can last for many years when exposed to the elements.
- Some types of bamboo have a durability of only a few months.

Durability: resistance to wear, long lasting.

Now test yourself

TESTED

1 Why is plywood a much stronger manufactured board than MDF? (2)
2 Name **one** advantage and **one** disadvantage of knots in pine. (4)
3 Explain **one** major difference between hardwood and softwood trees. (2)

1.13 Contexts which inform design outcomes

1.13.1 Discriminating between materials, components and manufacturing processes

REVISED

Properties may change depending on things such as temperature or the shape and direction of the material. Other properties will remain the same regardless of the environment or conditions.

Mechanical properties include:

- compressive strength – resistance to a pushing force
- tensile strength – resistance to a pulling force
- durability – resistance to wear, long lasting
- hardness – resistance to abrasion
- stiffness – resistance to bending and flexing
- elasticity – ability to stretch and return to shape
- impact resistance – resistance to knocks and impacts
- brittleness – hard but easily broken or shattered
- water resistance – resistance to water or moisture
- absorbency – ability to soak up moisture.

Manufacturing properties include:

- machinability – ability to be easily milled and drilled
- workability – ability to be placed and compacted
- ductility – ability to stretch, not break under stress or impact
- malleability – ability to be hammered, pressed or rolled without breaking.

Other properties include:

- acoustic – acoustic absorption
- chemical – corrosion resistance
- electrical – conductivity, resistance and so on
- thermal – conductivity, flash point and so on
- magnetic – attractive to magnets.

Typical mistake

Malleability and ductility are similar properties and often confused. Ductility relates to the ability of a material to stretch without damage. Malleability relates to the ability of a material to deform under compression.

Exam tip

By learning the properties of materials, you can understand why they have been used to make certain products. This will help you to answer questions in the exam about individual products and their fitness for purpose.

Advantages and disadvantages

- The choice of materials impacts the success of the final product.
- The advantages of a particular material may make it an obvious choice for a product, but the designer must also consider disadvantages, for example a thermosetting polymer may have the best properties for a product but it is not recyclable. Therefore, the designer may choose a thermoforming polymer with properties as close to the thermosetting one as possible.

Justification of the choice

- The designer must make informed choices based on these characteristics and other factors, such as:
  - cost – is there another material available that may cost less?
  - availability – is the material in good supply?
  - sustainability – is the material sustainable?
  - environmental impact – can the material be recycled?
  - social, cultural and moral considerations – will the use of this material have a negative impact?
- When designing an electronic system, the designer must consider the range of components available and decide on the most suitable type for the application.

Now test yourself

TESTED

1 State **four** important mechanical properties of materials. (4)
2 State **two** factors that can influence and affect the properties of materials. (2)

1.14 Environmental, social and economic challenges

1.14.1 Respect for different social, ethnic and economic groups

REVISED

Positive and negative impacts of products

- Products can have a negative impact on society that the designer may not have foreseen.
- For example, leaflets and flyers can be used to communicate useful information to people about services in the area, however this can lead to unwanted junk mail being delivered, urging people to buy things they don't want.
- While it is not always easy to predict the effects of a product, designers must try to consider the possible negative implications and decide whether the product is worthwhile.

Cultural awareness

- **Cultural awareness** is considering how different cultural groups may be affected by a product. Different cultures may view products in a totally different light depending on their beliefs, ideas and experience.
- A product designed for a particular market sector should not be offensive or upsetting to another.
- Designing new products that continue traditional processes, skills and materials can help preserve cultural identity.

Cultural awareness: considering how different cultural groups may be affected by a product.

Economic challenges

- The manufacturing processes, transportation, use and disposal of a product can affect the economic needs of people.
- The manufacture of a new product may create jobs in a factory or geographical area, boosting local people's income and improving the overall economy of the area.
- If the product is manufactured with modern CAM machines, it could lead to the loss of jobs and, in turn, affect the whole economy in that area.
- Many products are made abroad because labour costs are much cheaper in developing countries.
- Some workers in these countries are poorly paid, work in dangerous or unhealthy conditions, and work long hours without breaks or holidays.
- In some countries there is use of child labour, with children working in unsafe conditions because they have no choice.
- Designers have a responsibility to ensure that people are not being exploited in this way.

1.14.2 Appreciation of the environmental, social and economic issues

REVISED

- Designers should make sure their clients follow the guidelines for the fair treatment of workers set out by organisations such as the European Institute for Crime Prevention and Control, the Gangmasters and Labour Abuse Authority (GLAA) or the Alliance for Workers Against Repression Everywhere (AWARE).
- Manufacturers of fairtrade products ensure that their workers are treated and paid fairly, and that they do not work in dangerous or unhealthy conditions (see Section 1.2.3 Ethical perspectives on page 12).
- Carbon offsetting is when companies that use high amounts of energy or produce a lot of waste 'offset' this by setting up or donating funding to projects that help to reduce greenhouse gases.
- Products that can be disassembled easily allow the components to be reused instead of throwing away perfectly useable parts and manufacturing new ones. Making it easier to separate parts for recycling means that less materials are disposed of in landfill.

1.14.3 Green design

REVISED

- Social pressure to have the latest products and the influence of fashion and trends mean that people throw away rather than repair things. This is known as the **throwaway society**.
- **Green design** means using eco-friendly materials and construction practices that safeguard air, water and earth.
- **Eco-design** means designing sustainable products that will not harm the environment by considering the effects of the technology, processes and materials used.

Throwaway society: buying products for brief use then disregarding them or throwing them away with little thought of the environmental impact.

Green design: using eco-friendly materials and practices to design products.

Eco-design: designing sustainable products that have a minimal effect on the environment.

Stakeholder: a person other than the main user who comes into contact with or has an interest in the product.

1.14.4 Recycling and reusing materials or products

REVISED

- Choose materials that are sustainable, recyclable and non-toxic, and which do not require as much energy to process.
- Design products to last as long as possible, so there is less replacement of parts, and that can be fully recycled at the end of their useful life.
- Design products that function to the best of their potential.
- Consider the needs and wants of all the **stakeholders** involved.

Typical mistake

Sustainable products and materials are not the same as recyclable products and materials.

1.14.5 Human capability

REVISED

- Human capability is the range of functions that humans can carry out.
- Cognitive capability is the ability to process thoughts, learn new things and solve problems.
- Physical capability is the ability to perform physical tasks, such as walking, standing, balancing or gripping items.
- As we mature from children into adults, our cognitive and physical capabilities develop and increase.
- Old age, illness or other events in our life can have a huge effect on our human capability.

1.14.6 Cost of materials

REVISED

- Designers must try to keep material costs as low as possible without compromising the quality of the final product.
- Where materials are expensive, designers must look for cheaper alternatives with similar properties.
- Designers must consider whether a particular material is necessary or just for 'show'.

1.14.7 Manufacturing capability

REVISED

- Manufacturing capability is what a company or manufacturing plant is able to produce.
- Many companies have approved ways of producing a particular product.
- Manufacturing capability relates to the:
  - technological capability of the manufacturer – how up to date the equipment is